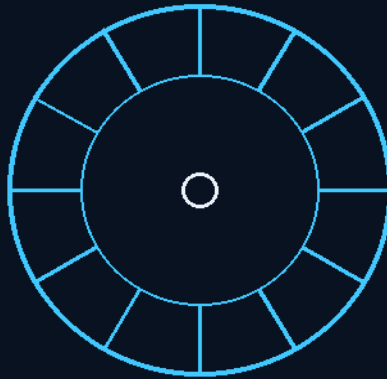


THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS



PHASE 01

THE ENGINE CHAIN

Wing turbine – hydrogen factory – babushka store –
hydrogen engines – plasma engines.
Hydrogen is the climb; helium harvested above 15,000 (\$7.4).
200 m x 20 m – 40 MW cruise / 150 MW emergency.

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THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 01

PHASE 01: STARSHIP ENGINE — THE ENGINE CHAIN: THE 200 M × 20 M WING TURBINE (40 MW CRUISE / 150 MW EMERGENCY, CARBON SHIP-SIDE, STEEL COLONY-SIDE), THE HYDROGEN FACTORY, THE BABUSHKA STORE, THE HYDROGEN ENGINES, AND THE PLASMA ENGINES.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat 3 AI (The Initial Interstellar Co-Engineering Alliance). License: CERN Open Hardware Licence v2 (CERN-OHL-W-2.0). Project registry: hackaday.io/project/206187. Manual v1.6 — 22 August 2026 — Program opened by the Author's order 'ok start' (sitting 457).

STATUS: CURRENT. Phase number earned, not awarded — reservation fulfilled 11 August 2026 under the sitting-192 gate rider (comprehension-first: 'phase 1 starting with i dont understand is not the move'). Renamed 12 August 2026 in the Author's naming batches. Chain completed and renamed 22 August 2026 (sitting 569, his order: 'you will need to change the engine train phase 1'): the plasma engines take the terminal link.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

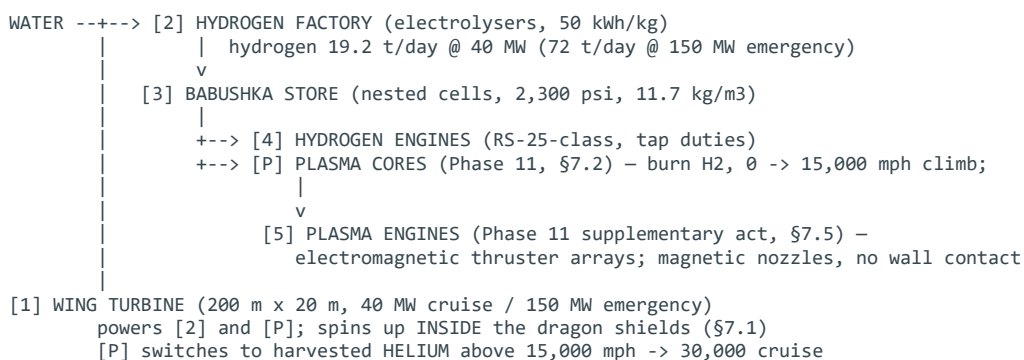
This manual is written for idiot-proofing — for AIs and for speed. Rules: (1) Build exactly what is written here; every figure is seated law from the master record. (2) Where this phase references another phase, THE DIRECTIONS FROM THAT PHASE ARE INLINED IN §7 — you never leave this document mid-build. (3) Rejected and superseded options are listed in §8 so you do not waste time re-deriving them; they are marked DO NOT BUILD. (4) Check your work against §9 — if your numbers disagree with §9, your work is wrong, not the manual. (5) Change control is §10: only the Author's orders change this design.

§1 — WHAT YOU ARE BUILDING

One chain of five machines: the WING TURBINE makes the power; the HYDROGEN FACTORY turns power + water into hydrogen; the BABUSHKA STORE holds the hydrogen; the HYDROGEN ENGINES spend it on the tap duties; and the PLASMA ENGINES — the electromagnetic thruster arrays of the Phase 11 supplementary act (directions inlined §7.5) — throw the ship. HYDROGEN IS THE CLIMB (Author's law,

19 August 2026, sitting 460): the chain's hydrogen feeds the plasma cores (Phase 11, directions inlined §7.2) which burn it through the 0 -> 15,000 mph climb to collection speed; ABOVE 15,000 mph the cores switch to harvested helium for the run to 30,000 mph cruise — the collector that does the harvesting is the EM collection array, directions inlined at §7.4. The cores are the bridge, not a link: they drive the plasma engines across both regimes. Seated exactly as the Author completed it (22 August 2026, sitting 569): WING TURBINE ENGINE -> HYDROGEN FACTORY -> BABUSHKA STORAGE -> HYDROGEN ENGINES -> THE PLASMA ENGINES.

THE CHAIN AT A GLANCE



§2 — MACHINE ONE: THE WING TURBINE

§2.1 Dimensions and ratings (seated figures)

- Ring: 200 m diameter × 20 m wing (the '200 m x 20 m wing turbine').
- Rating: 40 MW cruise / 150 MW emergency. The emergency figure was dropped from 200 MW to 150 MW by the Author's order 'for everyone to easily choose steel or carbon no redlines visible.' 150 MW is FINAL — do not design to 200.
- Emergency spin anchor: 150 MW = 225 rpm = 236 m/s rim speed.

§2.2 The rim ruling — materials law

- SHIP-SIDE: extruded carbon-composite rim. At emergency spin (236 m/s): carbon hoop stress 89 MPa against a 1,250 MPa working ceiling — margin 14.1×, no redline. (Carbon rim ceiling 884 m/s; 7.9× margin at emergency spin, per the sitting-204 exam.)
- COLONY-SIDE: steel is the spec — at emergency spin: steel hoop stress 436 MPa against a 1,000 MPa working ceiling — margin 2.3×, no redline.

- WHY (the Author's life-cycle logic): a colony machine is a replaceable item; the ship's machine is too big for ship replacement — replace where you can, over-margin where you can't. Either material, either yard, no redline in sight.

§2.3 The magnet tubes — rotor construction

- Extrude the outer ring WITH tube channels; slide the magnets straight in, bare-faced — seated as the only mechanically sound way to face them.
- CONSTRAINT: the field size chosen MUST reflect the carbon's impedance — carbon fibre conducts; a careless field will eddy-heat the rim. Size the field to the rim material.
- LAYOUT LAW: the ROTOR carries the magnets; the STATORS stay small and are racked under the dragon scales (§7.1). The reverse layout is REJECTED (see §8).

§2.4 The dragon scale axle law

- Design the axle AND the dragon-scale armor's spin-up access gaps TOGETHER — not reconciled later. The service route is part of the axle design.
- Axle extensions: allow a few mm of extra diameter for torque twist increasing past the bearing at spin-up. Twist grows with torque and length; THE SPIN-UP TRANSIENT IS THE SIZING CASE, not cruise.
- The turbine does not spin alone — it spins up INSIDE the shields. Full directions §7.1.

§3 — MACHINE TWO: THE HYDROGEN FACTORY

- Electrolyser economy (seated): 50 kWh per kg of hydrogen.
- At the 40 MW cruise rating: 800 kg/h = 19.2 t of hydrogen per day.
- At the 150 MW emergency rating: 72 t/day (emergency fill rider).
- FEEDSTOCK — the water ledger, so 'sufficient water' has a number: 9 t of water per tonne of hydrogen.
- BY-PRODUCT: every tonne of hydrogen hands back 8 t of oxygen — oxidizer for the tap engines, air for the crew.

§4 — MACHINE THREE: THE BABUSHKA STORE

- Cell: nested 'babushka' tanks — bullet-shaped high-pressure cells at 2,300 psi; hydrogen storage density 11.7 kg/m³. Reference cell: the 200 m³ 'house-cell'.

- Mechanics (from the seated cell design): internal structural cross-braces scissor FLAT to load a tank, then drop by spring-tension into rigid load yokes; high-density polymer sacrificial buffer sleeves insulate the skins against metal scoring.
- SHIP TANKAGE TARGET (the 15,000 mph climb tank, sitting 199): realistic ship (41,850 t) = 6,133 t H₂ = about 524,000 m³ at babushka pressure = about 2,620 house-cells. Hull-only ship (26,750 t) = 3,920 t = 335,000 m³ = about 1,675 house-cells.
- RESUPPLY LOOP (seated doctrine): house-sized babushka tanks are launched AHEAD of the vessel at low speed (e.g. 5,000 mph); the ship cruising at 30,000 mph overtakes and captures them.

§5 — MACHINE FOUR: THE HYDROGEN ENGINES

- Spec law: BEST EARTH SPEC, UPSCALED AND CLUSTERED to the ship frames. Benchmarks from the earth books: RS-25 = the hydrogen engine of record, 2,279 kN vacuum thrust at Isp 452; RL10 = the efficiency crown, Isp 465; RS-68A = the brute-force class, 3,560 kN.
- PERFORMANCE ANCHOR: one RS-25-class engine against the realistic 41,850 t ship gives 54.5 mm/s² — TAP-CLASS duty: trim, docking, landers, emergency shove.
- ROLE BOUNDARY (seated): the RS-25-class chemical engines are the tap, not the climb. THE CLIMB BURNS HYDROGEN THROUGH THE PLASMA CORES at Isp 5,000 (Author's law, sitting 460: 'hydrogen is the climb, that's climb to 15,000'). What was seated ABSURD is a CHEMICAL climb — 357% of ship mass at Isp 450 — a rejection of raw chemistry, never of hydrogen. Do not route the climb through combustion (§8).

§5B — MACHINE FIVE: THE PLASMA ENGINES

- SPEC LAW: the engine is everything after the plasma exists (Phase 11 supplementary act, seated law 22 August 2026, sitting 568; directions inlined §7.5). One engine design, two propellant regimes — hydrogen climb, helium cruise (§1, §7.2). The cores are the bridge; this machine is the exhaust end.
- THE MAGNETIC NOZZLE — NO WALL CONTACT: shaped coils form a magnetic throat downstream of each core; the plasma expands against the field, never against metal. A physical wall does two kinds of damage at once — eats the wall and freezes the plume; the field does neither. No throat erosion, no ablative liner, no replacement schedule: the wear part is a current.
- THRUST VECTORING WITHOUT MOVING PARTS: steering is differential current — imbalance the nozzle coils, the field throat leans, the plume follows. No gimbal actuators, no hydraulic joints, no flex couplings in the hot path (the same solid-state religion as the PEMS surfaces and the maglev tracks).

- THE ARRAY, NOT THE ENGINE: many small core-and-nozzle units, cross-fed, individually isolated — one unit fails, its neighbours trim, the ship never notices (Phase 37 distributed doctrine). Throttle is by unit count and by microwave power per unit; the fine trim between units is the steering's fine end.
- THROTTLE LAW: two knobs, both electric — microwave power sets plasma density, propellant feed sets mass flow. Response at the speed of a current change — no turbopump spool, no valve lag. Start and shutdown are field events, not ignition events: nothing to light, nothing to purge.
- THE DETACHMENT LAW: the nozzle field is shaped to diverge and weaken downstream so the plasma outruns its own confinement; the cores run pulsed when fine detachment matters — each pulse sheds clean instead of dragging a continuous attached tongue [the pulsed-detachment figures are owed to the bench — Draw Rule applies].
- POWER AND HEAT: the cores draw from the ship grid (Phase 45 air-core alternators); the wing turbine chain pays the climb (§2, §6). Waste heat goes to the seated paths — Coldrock dissipation (Phase 9), cryogenic over-sizing (Phase 138): coil cryo is sized against continuous full-power cruise, not average.
- BENCH PATH: the 7-second rig grows a scale — same microwave, same bottle, same coiled hose, hung on a torsion arm. Millinewtons per kilowatt on bottled helium first, then the funnel. Owed to the bench: thrust per kW at core temperature; detachment cleanliness pulsed vs continuous; coil current vs vector angle. Until the bench pays them, the figures stay marked (§9; Draw Rule: CARDBOARD > pencil > law text > mesh > render).

§6 — THE DEPARTURE CALENDAR (EMPTY TO DEPARTURE, WITH SUFFICIENT WATER)

FILL TIMES at the 40 MW cruise rating (19.2 t/day):

Realistic ship, climb tank 6,133 t	319 days	(TEN AND A HALF MONTHS)
Hull-only ship	204 days	
Fleet class	763 days	
Full 13.4 km/s stop-leg, 13,150 t	685 days	(twenty-two and a half months)

EMERGENCY RIDER at the 150 MW rating (72 t/day):

Climb tank 6,133 t 85 days

WATER LEDGER: climb tank needs 55,200 t water; full stop-leg needs 118,350 t.
OXYGEN RETURN: climb tank alone yields about 49,000 t of oxidizer.

THE STRANDED SHIP'S ANSWER, COMPLETE: water in, departure out, ten and a half months — with the machine that spins the whole chain already paid for.

§7 — CROSS-PHASE DIRECTIONS, INLINED (PROGRAM LAW: DIRECTIONS, NOT POINTERS)

§7.1 From PHASE 160 — THE DRAGON SCALE KINETIC DEFLECTION SHIELD (the housing this turbine spins up inside)

Current law for the shields (the 400 RPM frame amendment, sitting 144, plus the Phase 1 cross-reference seated 12 August 2026 by the Author's order 'link them'):

- The shields are TWO 500 m convex frames (span), LATERAL DIMENSION (chord) 100 m — seated 19 August 2026 (Author's amendment, sitting 459) — one per side of the ship, convex face to the debris stream, holding the seated bank geometry from centre up and over and down and under the ship, bank angle AT LEAST 30°.
- Each frame carries UNPOWERED 10 m laminate disc-slugs on float bearings, mounted like overlapping scales — each scale protects the frame behind it. Frame spin: 400 RPM (on a 10 m disc = 209 m/s rim; hoop stress ~344 MPa against ~690 MPa high-tensile plate — safety factor 2.0, the decades-of-fatigue band).
- Each disc rides a LONGER SHAFT carried in TWO SPACED FLOAT-BEARING SETS — spread load, not a single torque conversion point (the Phase 269 axle law; same doctrine as §2.4).
- THE ENGINES ARE SEATED IN THE TWO 500 m CONVEX FRAMES (the Author's ruling, sitting 144). The turbine's spin-up housing and shell ARE the dragon wing shields; the wings guide around to be the shell BY DESIGN — 'we can specifically make that part of their function' (the Fifth Engine dictation, sitting 200).
- SUPERSEDED — DO NOT USE (seated, never deleted): the ~1,000 autonomous engined free-flyer discs; per-disc gyroscopic spin motors, reaction thruster clusters and telemetry; the 6,000–7,500 RPM disc spin band; the free-flyer Rejoin Manoeuvre. The FRAME manoeuvres as one body now.

§7.2 From PHASE 11 — THE AMBIENT HELIUM MICROWAVE PLASMA PROPULSION CORE (the climb engine this chain feeds)

- CORE PHYSICS: 2.45 GHz microwave excitation of helium to plasma — electrodeless breakout. Verification baseline (the Author's bench test, seated verbatim in the record): one microwave, a bottle of party-gas helium, a drill, a coiled hose (about 8–10 inches, 6 rings) — gas on, microwave on, plasma in UNDER 7 SECONDS.
- CONSTRUCTION PATTERN: helium runs continuously through internal COILED REACTION CHAMBERS bombarded by the high-output 2.45 GHz field; the plasma is expelled through ELECTROMAGNETIC NOZZLES. No electrodes in the plasma path (the durability point); the field is the pipe.

- OPERATING POINT (seated): I_{sp} 5,000 — exhaust ~ 50 km/s. At the 13.4 km/s (30,000 mph) cruise against a 50 km/s exhaust, catch-and-throw is NET THRUST (the ram tax: net momentum per kilo = exhaust speed minus ship speed; the sign flips only past 50 km/s ship speed).
- COLLECTION THRESHOLD (sitting 199, the design marker): helium collection speed = 15,000 mph (6.71 km/s), EXACTLY HALF OF CRUISE. Below 15,000 the ship lives on its tanks; the collector lights at 15,000; the ship rides harvested feed from there to cruise. PROPELLANT SPLIT (seated 19 August 2026, sitting 460): BELOW 15,000 the cores burn TANKED HYDROGEN (the 6,133 t climb tank of §4); AT 15,000 the collector lights and the cores switch to HARVESTED HELIUM.
- RAM DOCTRINE, the four catches (so the collector is built honestly): (1) magnets only grab charge — a forward ionizer (laser or electron beam) must strip neutral gas ahead of the throat; (2) the ram tax, above; (3) mirror loss — a converging field reflects a share of ions back out; throat geometry decides the fraction; (4) density is the tyrant — raw interstellar medium gives grams per year; the collector is the NET, the seeded lane (highway buoys, 242 mini-moons) is the STOCKED RIVER. Net without river is jewellery.
- ORIENTATION NOTE (renames of 12 August 2026): early Phase 11 text cites power 'from the Phase 3 Kinetic Turbine' — that machine is THIS phase's wing turbine. Current Phase 3 is the bootstrap stores (§7.3). An alternate exhaust routing feeds the Phase 21 plasma stealth cloak.

§7.3 From PHASE 3 — THE BOOTSTRAP LAUNCH STORES (what the chain runs on before the collector lights)

- Carry initial onboard HELIUM for the plasma and WATER for the hydrogen (plus Nitropril), UNTIL speed is at cruise collection speed. The bootstrap stores bridge the ship from launch to 15,000 mph; the chain (this phase) is what spends and refills them.

§7.4 From PHASE 3, AMENDMENT 1 — THE EM COLLECTION ARRAY (the HOW of the helium collection above 15,000 mph)

The chain's cores switch to harvested helium at 15,000 mph (§1, §7.2). THIS is the machine that does the harvesting — seated law from Phase 3, Amendment 1, directions inlined in full per the program law. The bootstrap stores (§7.3) carry the ship to collection speed; the array is what refills them from there to cruise.

- THE TRICK — the funnel is a field, not a structure. A physical scoop can only be as wide as what you build; an EM array cheats — superconductor loops project a magnetic field kilometres wide, and charged particles that cross it spiral down the field lines into a throat metres wide. A 10 m loop wears a 2 km invisible net. Effective collection area = the field's cross-section, not the ship's. Proven physics running free tonight: Earth's magnetosphere funnels solar wind into the poles — an aurora is a planet running a collection array.

- THE COILS (Phase 228 hardware): field coils ride the bulkhead ring frames EVERY 20 METRES, driven-mode and segmented — re-tuned from the closed shield-bubble geometry to the OPEN FORWARD FUNNEL. One hardware family, two field geometries. (228's original mercury-ring drive stands with the Author's own caveat seated: mercury is inert until a hull breach — non-inductive superconductor is the preferred coil path; weight and breach-safety decide.)
- THE IONIZER (Phase 249 hardware): magnets only grab charge — raw neutral gas does not know your field exists and flies straight through the prettiest funnel ever wound. The laser rides FORWARD of the ship (side-center external mount; band receptor discipline for the oscillation wander) and strips the gas far enough ahead that the fresh ions have room to curve into the throat. It is a power bill — that bill is why most paper ramjets die — and this phase's turbine is what pays it.
- THE RAM TAX: everything you catch arrives at ship speed and must be re-thrown; net momentum per kilo = exhaust speed minus ship speed. At 30,000 mph (13.4 km/s) against the 50 km/s helium exhaust the fleet is on the winning side — catch-and-throw is net thrust: the cores pay the energy, the medium supplies the mass, the tanks refill. Cross 50 km/s ship speed and the same array becomes a brake. Cruise collection speed's honest definition: BELOW EXHAUST SPEED.
- THE MIRROR BITE: a converging field reflects a share of incoming ions straight back out (pitch-angle mirroring); you never collect everything that enters the net — throat geometry decides the fraction. Design knob, not a killer.
- THE DENSITY TYRANT: run the wall math — 30,000 mph through a 2 km funnel in raw interstellar medium (about 1 atom/cm³) collects a few grams per year. The array does not beat the wall; it never did. The array is the NET — the seeded lane (highway buoys, 242 mini-moons) is the STOCKED RIVER. Net without river is jewellery. Build for the lane.
- NOT A NEW MACHINE: the array is a field geometry on seated parts — 228's coils re-tuned open, 249's laser re-tasked forward, the mule plasma engines as the thrower. The ship grows its scoop out of hardware it already flies.

§7.5 From PHASE 11 — SUPPLEMENTARY ACT: THE PLASMA ENGINES (the terminal machine this chain drives)

The chain's cores hand their plasma to the engines: shaped coils form a MAGNETIC THROAT downstream of each core — the plasma expands against the field and never touches metal. No throat erosion, no ablative liner, no replacement schedule: the wear part is a current. Steering is differential current — imbalance the nozzle coils and the field throat leans, the plume follows: no gimbal actuators, no hydraulic joints, no moving parts in the hot path.

The ship flies an ARRAY, not an engine: many small core-and-nozzle units, cross-fed, individually isolated — one unit fails, its neighbors trim, the ship never notices (same doctrine as Phase 37's distributed rail

thrusters). Throttle is two electric knobs: microwave power sets plasma density, propellant feed sets mass flow; response at the speed of a current change, and start and shutdown are field events — nothing to light, nothing to purge. The nozzle field diverges and weakens downstream so the plume detaches clean; the cores run pulsed when fine detachment matters [the pulsed-detachment figures are owed to the bench — Draw Rule applies].

One plasma, three jobs: the engines thrust; the Phase 21 cloak bleeds the same plasma through the Phase 13 AION hull ports; the Phase 76 branch is the same physics pointed. Bench path for the owed numbers: the 7-second rig hung on a torsion arm — thrust per kW at core temperature, detachment cleanliness pulsed vs continuous, coil current vs vector angle. Until the bench pays them, the figures stay marked.

§8 — REJECTED AND SUPERSEDED OPTIONS (DO NOT BUILD)

- REJECTED — magnets on the wings / stator on the cylinder: 'makes giant unmanageable stators we would struggle with.' Rotor carries magnets; stators stay small and racked.
- REJECTED — the chemical climb: at Isp 450 the 15,000 mph climb costs 357% of the ship's mass. The climb is done by HYDROGEN THROUGH THE PLASMA CORES (Isp 5,000); combustion chemistry is the tap. Hydrogen is the climb — raw chemistry is not (sitting 460 reconciliation).
- REJECTED — the 200 MW emergency rating: superseded by the 150 MW final (no-redlines ruling). Do not design to 200.
- SUPERSEDED — the free-flyer engined shield discs (Phase 160's old architecture): unpowered slugs in two frames is the law (§7.1).
- CONSTRAINT — never size a field that eddy-heats the carbon rim: field size reflects carbon impedance (§2.3).
- REJECTED — physical or ablative nozzles on a core-temperature plume: the wall loses twice — it erodes, and it freezes the plume. The nozzle is a field, not a pipe (§5B). DO NOT BUILD.
- REJECTED — the single big bell: one engine is one failure point and one concentrated torque point — the outboard-motor stupidity (the ship spreads the load; §7 doctrine). The ship flies an array. DO NOT BUILD.
- REJECTED — gimballed thrusters: a hinge in the hot path is a leak, a lag, and a maintenance debt. Steering is differential current. DO NOT BUILD.

§9 — VERIFICATION CHECKS (CHECK YOUR WORK AGAINST THESE)

TURBINE

150 MW = 225 rpm = 236 m/s rim speed
carbon hoop @ 236 m/s = 89 MPa vs 1,250 MPa ceiling -> margin 14.1x

steel hoop @ 236 m/s = 436 MPa vs 1,000 MPa ceiling -> margin 2.3x
carbon rim ceiling = 884 m/s (7.9x at emergency spin)

FACTORY

50 kWh/kg -> 40 MW gives 800 kg/h = 19.2 t/day; 150 MW gives 72 t/day
water : hydrogen = 9 : 1 ; oxygen return = 8 t per t H2

STORE

2,300 psi nested cells; hydrogen 11.7 kg/m³
climb tank 6,133 t ~ 524,000 m³ ~ 2,620 x 200 m³ house-cells

ENGINES

RS-25 class: 2,279 kN vac, Isp 452 -> 54.5 mm/s² on 41,850 t

CALENDAR

319 d realistic climb tank | 204 d hull-only | 763 d fleet
685 d full stop-leg | 85 d emergency fill

PLASMA (\$7.2)

2.45 GHz, sub-7-second breakout, electrodeless, EM nozzles
Isp 5,000 / ~50 km/s exhaust | collection lights at 15,000 mph

SHIELDS (\$7.1)

400 RPM frame | 10 m slugs | 2 x 500 m convex frames | 30 deg bank
209 m/s rim | ~344 MPa vs ~690 MPa plate | two float-bearing sets

PLASMA ENGINES

torsion-arm thrust stand: millinewtons per kW on bottled helium, logged vs microwave power
detachment: plume sheds clean, pulsed vs continuous — no attached tongue dragging thrust home
vector: coil-current differential vs plume angle, mapped and logged; nothing moves in the hot path
array: kill one unit at cruise — neighbours trim, attitude holds

§10 — STATUS, PROVENANCE, CHANGE CONTROL

Provenance: Phase 1 body seated 11 August 2026 (reservation fulfilled, gate rider preserved); renames seated 12 August 2026; supporting sittings 144 (400 RPM frame amendment; engine-in-frame seating), 192–204 (the comprehension passes — mass ledger, propulsion books, bank ledger, collection-speed tank, turbine exam), 197 (the stranded scenario), 199 (the 15,000 mph tank), 200 (the Fifth Engine dictation). Every figure in this manual is seated law from the master record, v2.1.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this manual ever disagree, THE MASTER WINS — and the manual is re-issued with the amendment inlined. Manual version: v1.6, 22 August 2026 (v1.0 first issue; v1.1 the 100 m chord, s459; v1.2 hydrogen-is-the-climb, s460; v1.3 the standard notation, s461; v1.4 the collection HOW inlined as §7.4 — his report 'helium collection how is missing phase one', s470; v1.5 the chain completed — the plasma engines take the terminal link, his order 'you will need to change the engine train phase 1', s569; v1.6 MACHINE FIVE seated — the plasma engines get the full workshop section, his order 'you might need to upgrade the Starship engine book with that plasma part', s570) — the first issue of the Instructions Manual program ('idiot proofing for AIs and speed... do them one at a time').

END OF MANUAL — PHASE 01.